

RABBY ZIL JALAL KANON

Computer Science & Technology Student

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github.com/Rabbyziljalal | [LinkedIn Profile](#) | [Portfolio Website](#)

STUDY OBJECTIVE

Final-year Computer Science and Technology undergraduate from Bangladesh, currently studying at Nanjing University of Posts and Telecommunications, China, seeking admission to a graduate program in Computer Science / Artificial Intelligence. My interest in technology began with curiosity about how websites work and has grown into a comprehensive understanding of full-stack development, AI algorithms, and modern software engineering practices, demonstrated through independently designed and implemented technical projects spanning machine learning, computer graphics (OpenGL/GPU programming), and web application development. I am particularly motivated by building solutions that create a positive impact on people's lives, and I actively pursue continuous learning through new technologies and open-source contributions.

EDUCATION

Bachelor of Computer Science and Technology <i>Nanjing University of Posts and Telecommunications (NJUPT), China</i> Specialization: Software Engineering & Artificial Intelligence International Student Scholarship recipient	2023 – present
Higher Secondary Certificate (HSC), Science Group <i>Bangladesh</i> GPA: 5.00 / 5.00	2021
Secondary School Certificate (SSC), Science Group <i>Bangladesh</i> GPA: 5.00 / 5.00	2019

HONORS & SCHOLARSHIPS

International Student Scholarship — Nanjing University of Posts and Telecommunications (merit-based, awarded to international students)

RESEARCH

Evaluating the Generalization of Flow-Based VPN Traffic Classification under Leakage-Resistant Data Splits

Causal Domain Adaptation for Distinguishing Benign Operational Drift from Adversarial Process Drift in Cross-Plant Industrial Control Systems

TECHNICAL PROJECTS

Artificial Intelligence & Machine Learning

Feng Shui AI — Smart Urban Location Analysis System

Built an AI-powered urban-analysis system combining traditional Feng Shui principles with machine learning; trained a Random Forest model on AMap POI/road data and HydroSHEDS river-network data to score locations.

Mandarin Master

AI-based Chinese-learning platform covering HSK 1–4 syllabus; delivers vocabulary, grammar, and sentence-structure practice through interactive AI-powered lessons.

Taskademia

AI-powered smart learning media platform for organizing and delivering educational content.

Computer Graphics & Systems Programming

GPU Particle Simulation — Compute-Shader-Powered Particle Engine

Real-time simulation of 2 million particles using GPU compute shaders, with rainbow-spectrum coloring and additive blending.

Interactive 3D Candle with GPU-Based Fire Particle System

Realistic 3D candle simulation with a modeled stand and a GPU compute-shader-driven fire particle system, including gravity, additive blending, and a rotating emitter.

OpenGL Interactive Triangle Renderer with ImGui

Graphics application rendering a shaded triangle with custom vertex/fragment shaders and a real-time ImGui control panel.

TCP Factorial Calculator — Multi-Client Client-Server App

Multithreaded Java TCP server handling concurrent authenticated clients requesting factorial calculations, with server-side logging and response timing.

Hash Table Design & Implementation

Custom hash table data structure implemented from first principles to study collision handling and performance trade-offs.

Rock-Paper-Scissors — Java Swing Multiplayer Game

Two-player networked Rock-Paper-Scissors game built with a Java Swing GUI, using TCP sockets to synchronize moves and results between client and server in real time.

Full-Stack & Application Development

GlobalTalk — Explore Cultures, Discover Food, Talk to the World

Cross-platform app for global food discovery and travel community, connecting users across 195 countries via real-time chat, an AI travel/food assistant, and translation.

FlyNova — Full-Stack Online Flight Ticket Booking System

Full-stack flight-booking platform with real-time flight search, seat selection, online check-in with boarding passes, and role-based admin controls.

Online Exam Management System with Webcam Proctoring

Full-stack exam platform supporting exam creation, webcam-monitored exam-taking, answer uploads, admin grading, and result review.

KidBattle (Brain Tug)

Browser-based educational tug-of-war game for children aged 3–10; teams answer math/English questions to pull the rope, with adjustable difficulty and team customization.

Aesthetic Digital Clock

Real-time digital clock web app with custom theming, smooth animations, and a responsive, visually polished layout.

TECHNICAL SKILLS

- Advanced: Java, HTML, Algorithm Design, Problem Solving
- Proficient: C, C++, JavaScript, Python, Data Structures, Web Development
- Web & Frameworks: React, Node.js, Express, Spring Boot, Flutter, TypeScript, Dart
- AI/ML: Scikit-learn, Random Forest, TensorFlow, PyTorch (working knowledge)
- Graphics/Systems: OpenGL, GLFW, GLEW, GPU compute shaders, Java Swing
- Databases: MySQL, MongoDB
- Tools: Git, GitHub, Docker, Linux, AWS (basics)

LANGUAGES

Bengali | English | Mandarin Chinese (Working proficiency, studying in China since 2023)|Hindi